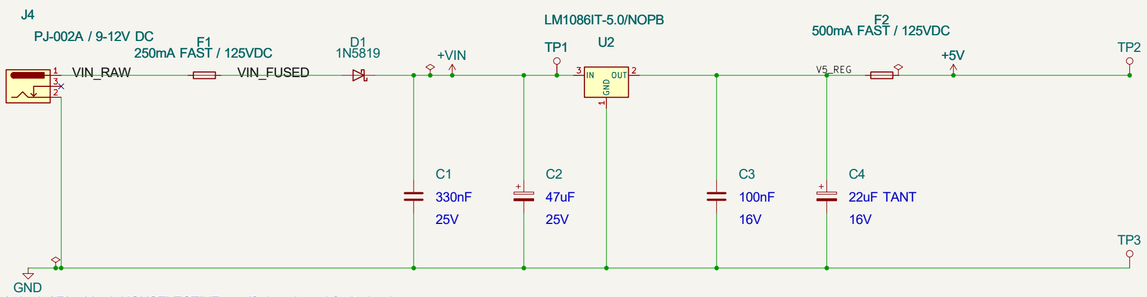
WEATHER GADGET — LCD DISPLAY SUBSYSTEM

Zander Boward | Team 202 | Revision J | 3 November 2026 | Newhaven 1602 LCD

ONE A2 SHEET | 5 V PIC / LCD logic / backlight | Three UART links

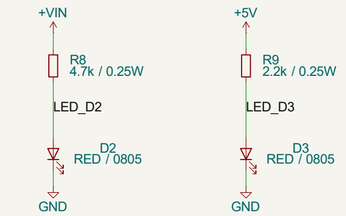
9–12 V INPUT / 5 V LINEAR REGULATOR

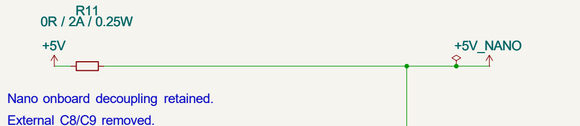


F1 250mA / F2 500mA NONSELECTIVE; verify inrush and fault clearing.

U2 LM1086: pin1=GND, pin2=OUT/TAB, pin3=IN. TAB IS LIVE 5V.

POWER INDICATORS





Nano onboard decoupling retained.

External C8/C9 removed.



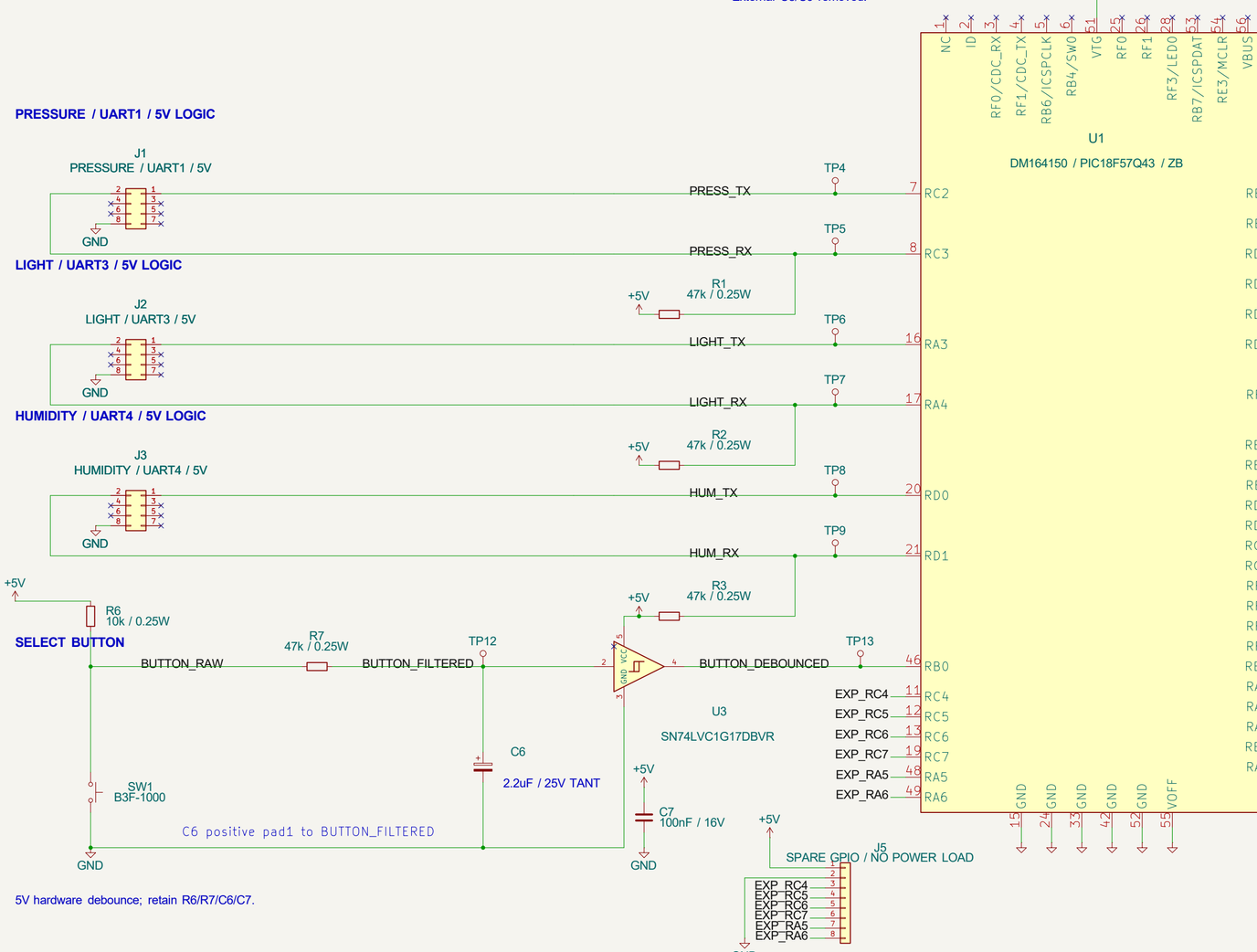
FB1: logic/backlight via pin2. Pins3/15/16 NC.

LCD supply hardwired via R12; signals can backfeed if disconnected.

PRESSURE / UART1 / 5V LOGIC

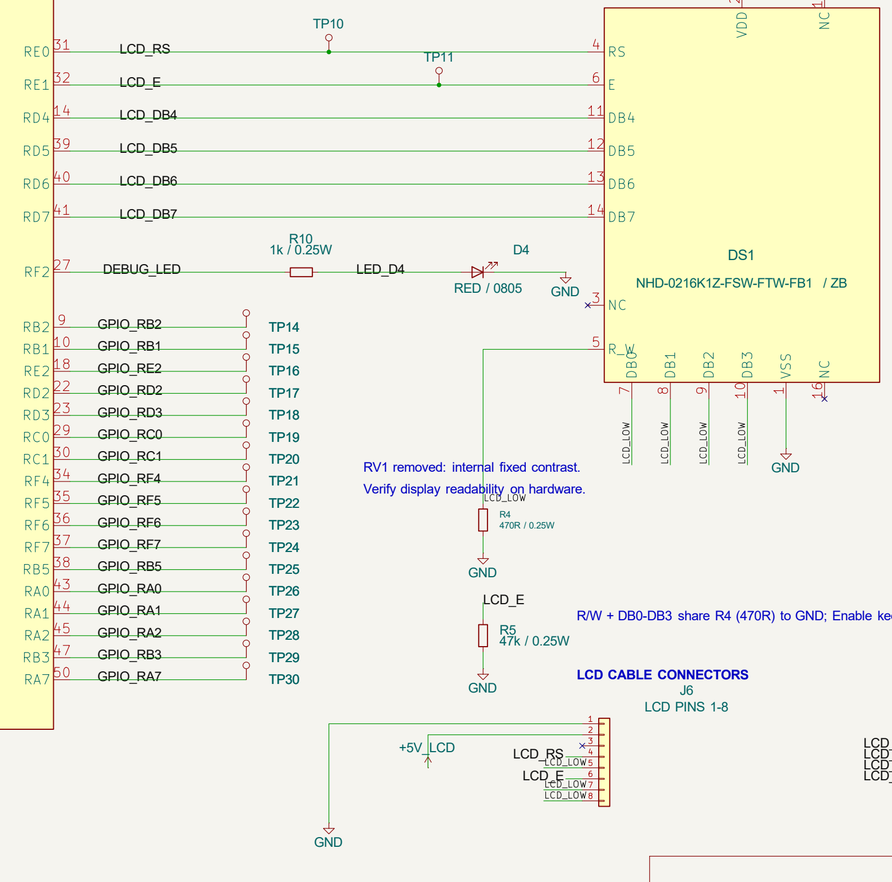
LIGHT / UART3 / 5V LOGIC

HUMIDITY / UART4 / 5V LOGIC



5V hardware debounce; retain R6/R7/C6/C7.

NEWHAVEN FB1 / FIXED CONTRAST / 5V



RV1 removed: internal fixed contrast.

Verify display readability on hardware.

R/W + DB0-DB3 share R4 (470R) to GND; Enable keeps separate R5.

LCD CABLE CONNECTORS

UART: pin 1 local TX -> peer RX; pin 2 local RX <- peer TX; pin 8 GND; pins 3-7 NC. No power exported.

Verify peer 5 V compatibility and cable continuity. VTG 51 = 5 V; VOFF 55 = GND; VBUS 56 NC.

ALL SOURCES OFF before changing cables. Disconnect powered peers for unpowered tests.

100mA load envelope; Nano provisional. Bench voltage/thermal/inrush checks PENDING.

REVIEW DRAFT: hardware qualification pending; 0–40C; no J5 powered load

FB1 fixed-contrast LCD; LM1086 5V; existing MCU/UART assignments retained

Team 202 | Zander Boward

Sheet: /

File: LCD-Board-Connected.kicad_sch

Title: Weather Gadget — LCD Display Subsystem

Size: A2	Date: 2026-11-03	Rev: J
KiCad E.D.A. 10.0.7		Id: 1/1